The qrime programme: what was actually achieved

A twin-prime analogue over $\mathbb{F}_q[x]$, attacked until the problem split into one finite-field inequality, one character-sum identity, and one machine-checked impossibility theorem. This document reports each of those pieces at its true status — proved, Lean-certified, numerically verified, or still open — and explains what the whole structure is worth.

Source: …/018 · addendum: …/019 | 106 round notes | 45 Lean modules | 111 check scripts

Harvested at round 106

— not over $\mathbb{Q}$, and not over $\mathbb{F}_q[x]$ either. It is a harvest of the programme's *map*: what is proved, what is computed, what is guessed, and where the single remaining gap sits. Speculative items are relabelled throughout; the sources call them “wild guesses”, and the change of word is deliberate.

ABSTRACT

The programme studies the twin-prime problem in the polynomial model: over a finite field $\mathbb{F}_q$, “primes” become monic irreducibles (*qrimes*) and “twin primes” become pairs `(f, f+2)` of irreducibles. In that model the twin count is a theorem (Pollack; Bary-Soroker–Stix), so the interest shifts: *which statistics can each tool see?*

Three results stand: (i) a **separation theorem** — the zeta function determines exactly the linear (one-point) statistics, and no function of it determines the twin count, proved with an explicit witness over $\mathbb{F}_7$ and formalised in Lean; (ii) a **finite-field cap** — for every quartic over $\mathbb{F}_q$, the reducible set has size at least `(q+1)/2`, proved unconditionally in the degenerate cases and reduced in the generic case to three standard theorems, with the algebraic links machine-checked; (iii) a **parity theorem** — every bounded-support divisor sum is periodic while the twin indicator is not, so no Selberg-type sieve weight equals the twin indicator, proved and Lean-checked. On the quantitative side, exact closed forms are obtained for the twin-qrime count at degrees 3 and 4 (including $A = (3q^2+2q+3)/16$ and $m = (a^2+1)/2$ for $q = a^2+b^2$), verified at 28 primes and identified with the quartic Jacobi sum $J(\chi_4,\chi_4)$. The entire remaining arithmetic is one twisted-Parseval identity: verified exactly at 80 primes with its

hypothesis sharpened to $q \equiv 1 \pmod 8$, and still unproved. The programme's own verdict is retained here: the tools it built map a barrier precisely; they do not cross it.

Keywords: function-field twin primes · qrimes · parity barrier · value sets of polynomials · character sums · Jacobi sums · Selberg sieve · Lean 4 formalisation · negative results

1	28	45	111	80
open identity, sharply stated	primes confirming the m-formula	Lean modules in the certified layer	executable check scripts	primes, exact F4 test

3	0
cited theorems behind the cap	proofs of the twin prime conjecture

1 The research

The twin prime conjecture asks whether n and $n+2$ are both prime for infinitely many n. The programme takes the standard move of transplanting the question into $\mathbb{F}_q[x]$, where "prime" means "monic irreducible". The transplant has a cost and a benefit: the *count* becomes provable (so the experiment can run), while the *method* that proves it — two-dimensional equidistribution, i.e. Chebotarev for the pair $(f,\ f+2)$ — is exactly what is missing over $\mathbb{Z}$.

1.1 The dictionary

The transposition that defines the programme.

$\mathbb{Z}$	$\mathbb{F}_Q[X]$
integers	monic polynomials
primes p	**qrimes** — monic irreducibles
size $\lvert n\rvert$	$q^{\deg}$
$p,\ p+2$	$f,\ f+2$ — **twin qrimes**
$\zeta(s) = \Sigma\ n^{-s}$	$1/(1 - q^{1-s})$
$\log\ \zeta$ coefficient	$q^n = \Sigma_{d\mid n}\ d\cdot I(d)$

Two counting functions carry everything. `I(n)` is the number of qrimes of degree `n`; `T(n)` — written `T(d;q,h)` when the shift and degree need emphasis — is the number of monic degree-`n` polynomials `f` such that both `f` and `f+h` are irreducible. `I` is **linear** data (a sum of one-point indicators); `T` is **bilinear** (a sum of products of two one-point indicators at a related pair). That distinction is the whole programme.

1.2 The three targets

After the early rounds the programme fixed three targets, chosen so that a proof of all three would give a complete, auditable account of the barrier. The synthesis at round 106 states them as follows (round106-final-synthesis-v2.md).

Target A — the `d = 4` cap

> `|(−Im g) ∪ Q| ≥ (q+1)/2` for every quartic `g` over `𝔽_q`.

Equivalently: for every quartic `g` and every constant `b`, at most `(q−1)/2` values of `b` make `g+b` reducible. This is a finite, self-contained statement about value sets of quartics — the part of the programme that stands on its own.

Target B — closed forms

Exact formulas for `T(d;q,h)` as a function of the shift class and `q`, and for the constants inside them. This is the quantitative heart: it says precisely how many twin qrimes there are, up to one integer.

Target C — the parity barrier

A theorem that the twin indicator lies outside the divisor-sum (Selberg-sieve) class, so that no sieve of that type can prove the twin prime conjecture. This is the "negative" target, and the programme treats it as the honest one.

1.3 Reading the labels

The single most useful thing a reader can carry into this material is the status ladder. The programme uses several different standards of evidence, and a claim's value depends entirely on which rung it is on.

Status labels used in this document and in the sources.

LABEL	MEANS	WHAT IT DOES *NOT* MEAN
LEAN	A Lean 4 / Mathlib theorem exists; the statement is machine-checked from the axioms.	Nothing about the informal reading of the statement, which still has to be checked by eye.
PROVED	A complete human argument exists, with no gaps the author is aware of.	It is not certified, and "modulo X" claims are not included here.
VERIFIED	Checked exactly by computer on a finite set of cases (often primes q).	It is *not* a proof; a pattern at 28 primes is strong evidence, nothing more.
OPEN	The statement is precisely formulated and unproved.	—
PROBE	A speculative mechanism offered to explain a pattern; the sources' own name for the same register is replaced here (§4).	It is not evidence of proof, and several probes in this programme have already been refuted.
REFUTED	Contradicted by an explicit computation or counterexample.	—

> **Rule of thumb used throughout this harvest.** A claim is reported at the weakest rung that honestly supports it. In particular: "proved modulo Ritt + Poincaré + the S_4 decomposition" is filed as *conditional*, not as proved; "verified at five primes" is filed as *verified*, not as a theorem — and now at 80 primes, still not a theorem; and a probe is never counted as a finding.

2 Findings, with proofs

Seven findings follow. Each states the result, gives the argument where the argument is short enough to be reproduced, and records the evidence and the status. Proofs of the machine-checked algebraic steps are given in the programme's own notation; the Lean identifiers are cited so the certificate can be located.

FINDING 1 The zeta function cannot see the twin count

PROVED LEAN Over $\mathbb{F}_q[x]$, the zeta function determines exactly the degree counts of a set of qrimes, and no function of those counts determines the twin count. This is the programme's cleanest structural result, and it is a statement about *information*, not about primes.

Lemma 1.1 (Euler–Möbius)

For a set S of monic polynomials with $Z(n) = \#\{f \in S : \deg f = n\}$ finite, define $\zeta_S(T) = \prod_{d\ge 1}(1-T^d)^{-Z(d)}$. Then

```
log ζ_S(T) = Σ_n (1/n) ( Σ_{d|n} d·Z(d) ) T^n ,
```

so ζ_S and $Z(S)$ determine each other. For the true world $S = \mathrm{Irr}_q$, $\zeta_S = 1/(1-qT)$, giving $q^n = \sum_{d|n} d\cdot I(d)$ and, by Möbius inversion, $I(n) = (1/n)\sum_{d|n}\mu(d)\,q^{n/d}$.

PROOF

Expand the logarithm of the Euler product: $\log\prod_d(1-T^d)^{-Z(d)} = \sum_d Z(d)\cdot(-\log(1-T^d)) = \sum_d Z(d)\sum_{k\ge1} T^{dk}/k$. Collect the coefficient of T^n: the pairs (d,k) with $dk = n$ contribute $\sum_{d|n} Z(d)/(n/d) = (1/n)\sum_{d|n} d\cdot Z(d)$. This is a bijection between $(Z(d))_{d\ge1}$ and the logarithmic derivative of ζ_S, hence between $Z(S)$ and ζ_S; the Möbius form follows by inverting $q^n = \sum_{d|n} d\cdot I(d)$. ∎

A curve's Frobenius eigenvalues likewise determine the point counts $\#C(\mathbb{F}_{q^n}) = \sum_{d|n} d\cdot Z_C(d)$, so the zeros also see only linear data. The decisive case is $\mathbb{P}^1$: its zeta is $1/((1-T)(1-qT))$, whose numerator is 1 — **there are no zeros at all**, yet the twin count is positive (over $\mathbb{F}_3$ at $n = 3$ there is 1 such pair; over $\mathbb{F}_7$ at $n = 2$ there are 7). A function with no zeros cannot be encoding a positive count that varies.

Theorem 1.2 (separation)

Quantify over *admissible worlds*: sets $S \subseteq \mathrm{Irr}_q$ with a degree function. Then there is no Φ such that $B(S) = \Phi(Z(S))$ for all admissible worlds, where $B(S) = \#\{f \in S : f+2 \in S\}$.

PROOF (EXPLICIT WITNESS OVER $\mathbb{F}_7$, DEGREE 2)

All eight degree-2 monic polynomials over $\mathbb{F}_7$ are irreducible. Take

```
S = { x²+1 ,  x²+3x+1 ,  x²+4x+1 ,  x²+3x+6 }      B(S) = 1
T = { x²+1 ,  x²+3x+1 ,  x²+4x+1 ,  x²+2    }      B(T) = 0
```

Both are sets of qrimes with the same degree profile $Z = (0,4,0,\ldots)$, hence the same zeta function. The twin counts differ: in S the only element whose shift lands back in the set is x^2+3x+6 (its shift is x^2+3x+1), while in T the shift of x^2+2 is x^2+4, which is reducible because $4 = 2^2$ is a square. Hence $B(S) \neq B(T)$ while $Z(S) = Z(T)$, so no Φ exists. ∎

Lean: `Separation.no_zeta_observable`, `Separation.separation_F7`, `Separation.witness_irreducible`.
Numerics: `checks/round3.py` §[C] tabulates the twin count on fibres of Z.

Two warnings attach to this theorem, and the programme is explicit about both. First, it concerns the *indicator*, not the count: it does not say twins are infinite, only that this channel of information is insufficient. Second, the theorem is about Φ depending on the zeta *alone*; a proof may use the pair structure directly — which is exactly what the function-field theory does, via Chebotarev for `(f, f+2)`.

What the zeta is missing is 2-dimensional equidistribution. Over $\mathbb{F}_q[x]$ it is available (the Frobenius gives a Galois action on the splitting field of $f\cdot(f+2)$); over $\mathbb{Z}$ there is no Frobenius and no Chebotarev, and the two-point equidistribution is precisely the twin prime conjecture. This is the structural reason a function-field proof exists and a $\mathbb{Z}$ proof does not follow from it.

FINDING 2 The d = 4 cap: a proved inequality about quartics

PROVED LEAN (algebraic links and finite base) For every quartic $g(x) = x^4+a_3x^3+a_2x^2+a_1x+b$ over an odd $\mathbb{F}_q$ and every b, the reducible set $R(g)$ satisfies

$$|R(g)| \geq (q+1)/2 , \qquad \text{equivalently} \qquad I(a) = q - |\mathrm{Im}(a) \cup Q(a)| \leq (q-1)/2 .$$

Here $I(a)$ counts the b making g_a+b reducible, $\mathrm{Im}(a)$ is the value set of $g_a = x^4+a_3x^3+a_2x^2+a_1x$, and $Q(a)$ is the set of b for which g_a+b has a quadratic factor. The identity $I = q - |\mathrm{Im} \cup Q|$ is the observation that a quartic is reducible exactly when it has a linear or a quadratic factor.

The proof is a four-step reduction. Steps (1)–(3) and the finite base are machine-checked; step (4) invokes three standard theorems.

(1) Translation to the depressed quartic

The substitution $x \mapsto x - a_3/4$ is the algebra automorphism `algEquivAevalXAddC (-a₃/4)` of $\mathbb{F}_q[x]$, hence preserves irreducibility and permutes the family $\{g+b\}_b$. So it suffices to treat $g = x^4+a_2x^2+a_1x$. Lean: `Shift.irreducible_shift`.

(2) The reducible set is a union

Factor a reducible quartic as $(x^2+px+q)(x^2-px+s)$ — always possible for a depressed quartic whose quadratic factors have no cubic terms. Matching coefficients gives

$$s + q - p^2 = a_2 , \qquad p(s - q) = a_1 , \qquad qs = b .$$

PROOF OF THE UNION DECOMPOSITION

If $p \neq 0$, the second relation gives $s = q + a_1/p$; adding $2q = a_2 + p^2 - a_1/p$ gives $q = (a_2+p^2-a_1/p)/2$, $s = (a_2+p^2+a_1/p)/2$, and therefore $b = qs = ((a_2+p^2)^2 - (a_1/p)^2)/4$. If $p = 0$, the second relation forces $a_1 = 0$ and gives $b = q(a_2-q)$. Finally the linear-factor case gives $b = -g(x)$. Hence $R = (-\mathrm{Im}\ g) \cup Q$ with $Q = Q_{p\neq 0} \cup (a_1=0\ ?\ \{q(a_2-q)\} : \varnothing)$. ∎

Verified for all quartics at $q = 11,13,29,37$. Lean: `Factorization.quadratic_factor`, `Factorization.even_factor`.

(3) The composition case $a_1 = 0$: the cap holds exactly

Here $g = x^4+a_2x^2 = h(x^2)$. Put $q_0 = x^2+a_2$. Then for every x,

$$-g(x) = -x^2(x^2+a_2) = q_0\cdot(a_2 - q_0) \in Q_0\ ,$$

so $-\mathrm{Im}\ g \subseteq Q$ and $R = Q$. Lean: `Composition.neg_image_mem_Q`. The size of Q splits into two cases.

- $a_2 = 0$: $Q = -\text{squares} \cup (\text{fourth powers})/4$. If $q \equiv 1 \pmod 4$ then -1 is a square, so $Q = \text{squares}$ and $|Q| = (q+1)/2$; if $q \equiv 3 \pmod 4$ the fourth powers coincide with the squares and $Q = \mathbb{F}_q$. Lean: `NegSquare.neg_sq_mem_sq`.
- $a_2 \neq 0$:

  $$|Q| = |Q_0| + |Q_1 \setminus Q_0| = (q+1)/2 + \lfloor (q-1)/4 \rfloor = q - \lfloor (q+2)/4 \rfloor\ .$$

  The first term is the quadratic image, of size $(q+1)/2$. The new part is an orbit count: the involution $\sigma(u) = -u-2a_2$ preserves the value $((a_2+u)/2)^2$, and its character argument $-(2a_2+\sigma(u)) = u$ never has value -1 when u is a square, so the count of remaining square classes is $(N-\Sigma)/2 - \frac{1}{2}\cdot 1_{\chi(-2a_2)=1} = \lfloor (q-1)/4 \rfloor$, with the character sum $\Sigma = -(\chi(-1)+\chi(-2a_2))/2$. Lean: `QSize.size_identity`, `QOrbit.sigma_y`, `QOrbit.sigma_arg`.

(4) The generic case $a_1 \neq 0$

This is where the three cited theorems enter. The quartic is not a composition (else $a_1 = 0$), so by **Ritt**, in the form of Guralnick–Saxl, its monodromy is *primitive*. The primitive transitive subgroups of S_4 are A_4 and S_4, with fibre-size ratios $3/4$ and $5/8$; hence $|\mathrm{Im}|/q \to \geq 5/8$. In the S_4 case the indicator $1_{k\geq 1}$ is expanded in the class functions of S_4:

$$1_{k\geq 1} = (5/8)\cdot 1 + (1/8)\cdot \mathrm{sign} + (3/8)\cdot \mathrm{std} - (1/8)\cdot(\mathrm{std}\otimes\mathrm{sign}) - (1/4)\cdot\chi_2 \ ,$$

(here χ_2 is the two-dimensional irreducible of S_4, not the quadratic character of $\mathbb{F}_q$). Since $\Sigma_y \chi_{\mathrm{std}} = \Sigma_y (N(y)-1) = q-q = 0$, the three non-trivial class sums give

$$|\mathrm{Im}\ g| = (5/8)q + (1/8)\left(S_{\mathrm{sign}} - S_{\mathrm{std}\otimes\mathrm{sign}} \right) + O(1) \ .$$

Both sums are traces of abelian varieties. $S_{\mathrm{sign}} = \Sigma_y \chi_2(\Delta(y))$ with Δ the cubic discriminant, which equals $-a_E$ for the elliptic curve $E : t^2 = \Delta(y)$. The pullback identity $\Delta(g(x)) = g'(x)^2 \cdot S(x)$ with

$$S(x) = -16x^6 - 32a_2x^4 - 40a_1x^3 - 20a_2^2x^2 - 36a_1a_2x - 27a_1^2 - 4a_2^3$$

makes $S_{\mathrm{std}\otimes\mathrm{sign}} = SC - S_{\mathrm{sign}}$ with $SC = \Sigma_x \chi_2(S(x))$. The genus-2 curve $C : s^2 = S(x)$ maps to E by $(x,s) \mapsto (g(x),\ g'(x)s)$, so by **Poincaré** reducibility $\mathrm{Jac}(C) \sim E \times E'$ and $S_{\mathrm{std}\otimes\mathrm{sign}} = \pm 1 - a_{E'}$. The **Weil bound** $|a_E| \leq 2\sqrt{q}$ then gives

$$|S_{\mathrm{sign}}| \leq 2\sqrt{q} \ , \qquad |S_{\mathrm{std}\otimes\mathrm{sign}}| \leq 2\sqrt{q} + 1 \ , \qquad |\mathrm{Im}\ g| \geq (5/8)q - (1/2)\sqrt{q} - O(1) \ .$$

Finally $(5/8)q - (1/2)\sqrt{q} - 2 \geq (q+1)/2$ exactly when $q - 4\sqrt{q} - 20 \geq 0$, i.e. for $q \geq 49$. The range $q < 49$ is discharged finitely: $q = 5,\ldots,23$ by the Lean `decide` theorems `SmallCap.cap_*`, and $q = 29,31,37,41,43,47$ by Python checks.

> **Status.** Steps (1)–(3), the finite base, and the orbit step are Lean-certified. Step (4) is a correct-in-outline argument that depends on Ritt/Guralnick–Saxl, Poincaré reducibility, the Weil bound, and the S_4 class-function decomposition. *None of those four is formalised*, and the synthesis itself lists their formalisation as the remaining work for Target A. So the honest reading is: **the cap is proved modulo three standard theorems plus one unformalised character-theoretic decomposition.** It is the strongest result in the programme, and it is not yet a certificate.

Sharpness and equality cases

The bound is attained, in two different ways, and the equality analysis is what suggested the "compensation" reading:

Equality cases of the cap (from the round-14 tabulation).

Q	MAX I	(Q−1)/2	MIN |IM ∪ Q|	(Q+1)/2
5	2	2	3	3
7	3	3	4	4
11	5	5	6	6
13	6	6	7	7

At $q = 11$ equality comes from $|\mathrm{Im}| = (q+1)/2$ with $Q \subseteq \mathrm{Im}$; at $q = 13$ from $|\mathrm{Im}| = 4$ with $|Q \setminus \mathrm{Im}| = 3$, the quadratic condition filling the gap. In both cases the value set alone would not suffice: Q is doing work. That is the mechanism the cap encodes, and it is a genuinely new finite-field phenomenon relative to the linear statistic.

FINDING 3 Closed forms for the twin-qrime count

PROVED VERIFIED The twin count has an exact structural description at $d = 3, 4$: it is constant on d-th power classes, its amplitudes are a d-point discrete Fourier transform of integer class counts, and two of its constants have closed forms.

(a) Scaling and power classes

PROVED The substitution $f(x) = a^d \cdot g(x/a)$ is a bijection on monic degree-d polynomials with $f + a^d h = a^d(g(x/a) + h)$, and it preserves irreducibility. Hence

$$T(d;\, q,\, a^d h) = T(d;\, q,\, h) \qquad \text{and } T \text{ is constant on } d\text{-th power classes.}$$

So the shift variable lives in $\mathbb{F}_q^\times/(\mathbb{F}_q^\times)^d$, a cyclic group of order d when $d \mid q-1$, and T is a function on $\mathbb{Z}/d$.

(b) The amplitudes are a DFT

PROVED Writing $T_k = T(d;q,u_k)$ for a choice of class representatives u_k, and $A_c = \Sigma_k\ T_k\ \zeta_d^{ck}$ for the d-point DFT,

$$d \cdot T_k = \Sigma_{c=0}^{d-1} A_c\, \zeta_d^{-ck}\,, \qquad c_4 = (1/4)[(A_0 - A_2) - i(A_1 - A_3)]\,.$$

The class counts T_k are integers, so each A_c lies in $\mathbb{Z}[\zeta_d]$; at $d = 4$ the assembled amplitude satisfies the parity integrality $c_4/(2q) \in \mathbb{Z}[i]$, which is proved. Lean: `TwinEven`.

(c) Divisibility by q

PROVED LEAN For $d < q$, $q \mid T(d;q,h)$.

PROOF

Let $\sigma_a : f(X) \mapsto f(X+a)$ for $a \in \mathbb{F}_q$. This preserves monicity, degree, irreducibility, and the pair condition $(f+h)(X+a) = f(X+a)+h$, so it acts on the set counted by T. The action is free when $d < q$: a nonzero translation-invariant polynomial has degree at least q, the minimal example being $X^q - X$. A finite set carrying a free action of a group of order q has cardinality divisible by q. ∎

The freeness hypothesis is sharp: at $d = q = 5$, $T(5;5,h) = 98$ for every $h \neq 0$, and $5 \nmid 98$, because $X^5 - X$ is translation-invariant. Verified for $d = 2,\ldots,6$ and every odd prime $q = 5,\ldots,31$ with $d < q$; no failures. Lean: the abstract engine is `FreeOrbit.card_dvd_card_of_free_action` (orbit–stabiliser plus the class formula).

Sourcing note. The round-91 note paraphrases the action as $t\cdot(f,g) = (f+t,\ g+t)$, i.e. translation of the *value*. That operation does **not** preserve irreducibility (e.g. x^4+b is reducible for about half the b). The preserved operation, and the one the round-23 argument actually uses, is translation of the *variable*, $f(X) \mapsto f(X+a)$. The Lean file proves the abstract orbit-counting lemma, which is correct and reusable; the application step is the round-23 argument.

(d) The even symmetry

PROVED LEAN $T(d;q,h) = T(d;q,-h)$.

PROOF

For $d \geq 1$ the map $f \mapsto f+h$ is a bijection of the monic degree-d polynomials (adding a constant preserves monicity and degree). It carries the property "f and $f+h$ both irreducible" to "$g = f+h$ and $g-h = f$ both irreducible", which is exactly the defining property of $T(d;q,-h)$. Hence the two counts agree. ∎

Lean: `TwinEven` (including `add_right_injective'`).

(e) The even-quartic irreducibility set, exactly

PROVED This is the cleanest complete description in the programme.

> $x^4 + a_2x^2 + b$ irreducible $\iff$ $\chi_2(b) = -1$ and $\chi_2(a_2^2-4b) = -1$.

PROOF

Write $f = h(x^2)$ with $h(y) = y^2+a_2y+b$. (i) h is irreducible iff its discriminant is a non-square, $\chi_2(a_2^2-4b) = -1$. (ii) If h is irreducible with root $r \in \mathbb{F}_{q^2} \setminus \mathbb{F}_q$, then $f = h(x^2)$ is irreducible iff r is not a square in $\mathbb{F}_{q^2}$. For $r \neq 0$, both "r is a square in $\mathbb{F}_{q^2}$" and "$N(r) = b$ is a square in $\mathbb{F}_q$" are the single condition $r^{(q^2-1)/2} = 1$, so they coincide. (iii) If h splits, f factors as two quadratics and is reducible. ∎

Counting with $1_{\chi_2=-1} = (1-\chi_2)/2$,

$$|B(a_2)| = (1/4)[\, q + \Sigma_b \chi_2(b)\chi_2(a_2^2-4b)\,] ,$$
$$|B(0)| = (q-1)/2 , \qquad |B(a_2)| = (q-1)/4 \quad (a_2 \neq 0) ,$$

since for $a_2 \neq 0$ the standard quadratic sum gives $\Sigma_b\chi_2(b)\chi_2(x-b) = -1$ at the nonzero square $x = a_2^2/4$. Verified at $q = 17,41,73$ for every a_2.

(f) The constant A

PROVED With the sizes above, Parseval gives the closed form

$$\Sigma_{a_2}|B(a_2)| = (q^2-1)/4 , \qquad S_0 = (q-1)^2/4 + (q-1)^3/16 ,$$
$$\Sigma_{c\neq0}\, S_c = (q-1)A = q(q^2-1)/4 - S_0 ,$$
$$A = (3q^2 + 2q + 3)/16 .$$

The step that makes it work is $\Sigma_{c\neq0}\chi_2(c) = 0$: the B-term drops out of the class sum. Verified exactly at seven primes ($q = 17,41,73,89,97,113,137$).

(g) The integer m, and the Jacobi sum

VERIFIED AT 28 PRIMES Write $q = a^2 + b^2$ with a odd and $b \equiv 0 \pmod 4$ — the unique such representation when $q \equiv 1 \pmod 8$, which is the case for all the data below. Then

$$2B = m\cdot\sqrt{q} , \qquad m = (a^2 + 1)/2 = (|\mathrm{Re}\ J(\chi_4,\chi_4)|^2 + 1)/2 ,$$

m against the representation of q as a sum of two squares.

Q	A^2+B^2	M	$(A^2+1)/2$	RE $J(X_4,X_4)$
17	1^2+4^2	1	1	−1
41	5^2+4^2	13	13	−5
73	3^2+8^2	5	5	3
89	5^2+8^2	13	13	5
97	9^2+4^2	41	41	−9
233	13^2+8^2	85	85	13

The identification matters because it converts an isolated arithmetic accident into a statement about a classical object: the quartic Jacobi sum satisfies $J(\chi_4,\chi_4) = \pm(a+bi)$, so m is a shifted square of its real part. One structural consequence: at $q = 41$ one has $m = 13 > 2\sqrt{41} \approx 12.8$, so m is *not* a single Frobenius trace; it must be a sum.

(h) The parity law, explicit

PROVED (from the decomposition) VERIFIED EXACTLY at seven primes.

$$E(h) = (q^2 - 2q - 7)/16 + ((a^2 + 1)/4)\cdot\chi_2(2h)$$
$$T(h) \equiv E(h) \pmod 2 .$$

DERIVATION

With $S_c = A + B\chi_2(c)$, $A = (3q^2+2q+3)/16$, $B = m\sqrt{q}/2 = (a^2+1)\sqrt{q}/4$, and $S_0 - A = q(q^2-2q-7)/16$, one has

$$E(h) = (S_0 - A)/q + (m/2)\cdot\chi_2(-h) = (q^2-2q-7)/16 + ((a^2+1)/4)\cdot\chi_2(2h) .$$

The two displayed forms agree because $q \equiv 1 \pmod 8$ makes 2 a quadratic residue, so $\chi_2(2h) = \chi_2(-h)$. ∎

All the m and E data above are at primes $q \equiv 1 \pmod 8$; the closed form $m = (a^2+1)/2$ presupposes the representation with a odd and $b \equiv 0 \bmod 4$ that this congruence guarantees. Stating the hypothesis matters: without it the two-square representation is not canonical.

FINDING 4 The single remaining identity

VERIFIED EXACTLY AT 80 PRIMES OPEN Everything in Target B reduces to one identity about a quadratic exponential sum. Define

$$K(A,c) = \Sigma_w \chi_2(A^2 - w^2)\, \zeta^{-cw/4}\ , \qquad H(a,c) = \zeta^{-ca^2/8}\cdot K(A,c)\ , \quad A = a^2/2\ .$$

K is real, and the two sums that carry the arithmetic are

$$\Sigma_A\, |K(A,c)|^2 = q^2 - 2q \qquad \text{(clean, independent of } c\text{)}$$
$$\Sigma_A\, \chi_2(A)\cdot|K(A,c)|^2 = 2\chi_2(c)\cdot\sqrt{q}\cdot \mathrm{Re}\, J(\chi_4,\chi_4)^2 \qquad \text{(the last identity)}$$

The first is an orthogonality statement — K is, up to a Gauss factor, the Fourier transform of $\chi_2(A^2-w^2)$, and Parseval gives q^2 minus the two boundary terms. The second is the whole arithmetic content of the programme, and it is the one thing that is not proved. Its consequence is exact: since

$$\Sigma_a\, |H(a,c)|^2 = |K(0)|^2 + 2\Sigma_{A \text{ square} \neq 0}|K|^2 = q^2 - 2q + 2\chi_2(c)\sqrt{q}\cdot \mathrm{Re}\, J(\chi_4,\chi_4)^2\ ,$$

the identity would make $\Sigma_a|H|^2$ — and hence the whole c_4 chain and the parity law — unconditional.

Two stages of the reduction

Rounds 99–105 localise the difficulty twice, and both steps are worth recording because they are proved. First, expanding $|T_2+T_3-T_4|^2$ and summing over a_2:

$$\mathrm{bulk_c} = (1/16)[\ 2q^2 + \Sigma|H|^2 + 2\chi_2(-1)\chi_2(-c)q\sqrt{q} + 4q\ ]\ ,$$
$$\Sigma_{a_2}|H(a_2,c)|^2 = q^2 - 2q - 2\chi_2(c)q\sqrt{q} + 4a^2\chi_2(c)\sqrt{q}\ ,$$
$$\mathrm{bulk_c} = (3q^2 + 2q)/16 + (a^2\sqrt{q}/4)\cdot\chi_2(c)\ .$$

So the entire a^2-dependence of m lives in the single sum $\Sigma_{a_2}|H(a_2,c)|^2$; every other term is a rational polynomial in q. That is a real theorem about where the arithmetic is hiding, and it is the reason the last gap can be stated as one line.

The falsification test, completed (019)

80 PRIMES, EXACT The round-106 note left a cheap and sharp test: one more prime. In 019 the test was run exhaustively, and — more importantly — *exactly*. Both sides of the identity are elements of the cyclotomic ring $\mathbb{Z}[\zeta_q]$: each is written as an integer vector of length q, and equality is decided by reduction modulo Φ_q. No floating-point tolerance enters the verdict, so a pass is a proof *for that prime*, not a numerical coincidence.

```
tested :  every prime q ≡ 1 (mod 4), q ≤ 1000   (80 primes),  c ∈ {1,2,3}
method :  both sides as integer vectors in ℤ[ζ_q], compared modulo Φ_q
runtime:  3.4 s
```

Representative rows. “strict” is the identity as stated in round 105; “general” is the corrected form below.

Q	Q MOD 8	TWO-SQUARE REP.	A^2-B^2	$2X_2(2)(A^2-B^2)$	STRICT	GENERAL
13	5	3^2+2^2	5	-10	FAILS	HOLDS
17	1	1^2+4^2	-15	-30	HOLDS	HOLDS
41	1	5^2+4^2	9	18	HOLDS	HOLDS
73	1	3^2+8^2	-55	-110	HOLDS	HOLDS
89	1	5^2+8^2	-39	-78	HOLDS	HOLDS
97	1	9^2+4^2	65	130	HOLDS	HOLDS
101	5	1^2+10^2	-99	198	FAILS	HOLDS
401	1	1^2+20^2	-399	-798	HOLDS	HOLDS
937	1	19^2+24^2	-215	-430	HOLDS	HOLDS
997	5	31^2+6^2	925	-1850	FAILS	HOLDS

Summary over all 80 primes.

FORM	HOLDS	VERDICT
strict: $2\chi_2(c)\sqrt{q}\ \mathrm{Re}\ J^2$	37 / 80	holds at **every** $q \equiv 1 \pmod 8$ (37/37) and at **no** $q \equiv 5 \pmod 8$ (0/43)
general: $2\chi_2(2c)\sqrt{q}\ \mathrm{Re}\ J^2$	80 / 80	holds everywhere, both classes
untwisted: $\Sigma_A\lvert K\rvert^2 = q^2-2q$	80 / 80	holds everywhere, as expected

Two things came out of the test, and both sharpen the target rather than close it.

1. **The stated hypothesis was too weak.** The identity does not hold for all $q \equiv 1 \pmod 4$: it fails at $q = 13, 29, 37, 53, 61, \ldots$ — every prime $\equiv 5 \pmod 8$. The measured boundary is *exactly* $q \equiv 1 \pmod 8$, which is the same as $\chi_2(2) = 1$. The programme's “in practice $q \equiv 1 \bmod 8$” is therefore not an artefact of the data: it is the hypothesis.

2. **The failure has a one-character fix.** At every prime tested,

> $$\Sigma_A \chi_2(A)\cdot|K(A,c)|^2 = 2\chi_2(2c)\cdot\sqrt{q}\cdot\mathrm{Re}\, J(\chi_4,\chi_4)^2 ,$$

which reduces to the stated form precisely when $\chi_2(2) = 1$. This is a sharper and uniform version of the identity, and it is what a proof should now aim at.

The test therefore neither proves nor refutes the identity: it *locates* it. No counterexample was found to the corrected form in 80 primes; the original form is dead outside $q \equiv 1 \pmod 8$. Supporting material in 019: results.json (all rows), falsification_test.py (the runner), test_twisted_parseval.py (14 unit tests), sympy_crosscheck.py (independent sympy/mpmath path at 50 digits).

What is certified, and what is not

The one purely combinatorial step in the derivation is that "sum over the squares" is "sum over everything plus the χ_2-weighted sum". That step is now a Lean theorem, for an arbitrary complex-valued function on $\mathbb{Z}/p\mathbb{Z}$:

```
theorem Twist.twisted_parseval_step (f : ZMod p → ℂ) (h0 : f 0 = 0) :
    2 * ∑ a, (if IsSquare a then f a else 0)
      = ∑ a, f a + ∑ a, (quadraticChar (ZMod p) a : ℂ) * f a
```

together with the pointwise form $2\cdot 1_{\text{square}} = 1 + \chi_2$ and the sanity corollary that $\mathbb{Z}/p\mathbb{Z}$ has $(p+1)/2$ squares. All five declarations are elaborated by `019/lean/Twist.lean` with only the standard axioms (`propext`, `Classical.choice`, `Quot.sound`).

What this certifies is that no arithmetic is hidden in the passage between the two sides — the identity's content is entirely in the character sum. What it does *not* certify is the character sum itself, which remains open. The green Lean file is a statement about the *reduction*, not about the identity.

What would close it

The programme's own probe (Probe 47, §4) is that the identity is the *twisted Parseval* for K: the untwisted sum is q^2-2q, and twisting by χ_2 replaces the trivial character by χ_2, whose Fourier transform is the Jacobi sum. That is a plausible mechanism, not a proof. Concretely, a proof would go through a Salié/Kloosterman evaluation of $\Sigma_{a_2}\chi_2(a_2^2-4v)\chi_2(a_2^2-4w)$ after completing the square — now with the $\chi_2(2)$ factor of the corrected form accounted for.

FINDING 5 The parity theorem: the twin indicator is not a divisor sum

PROVED LEAN This is the negative result that the programme treats as its honest core.

Lemma 5.1 (periodicity)

Let S be a finite set of divisors, all dividing L. Then for every $g : \mathbb{N} \to \mathbb{R}$,

$$\Sigma_{d\in S,\ d\mid n} g(d) = \Sigma_{d\in S,\ d\mid n+L} g(d) .$$

PROOF

For $d \in S$ we have $d \mid L$, so $n+L \equiv n \pmod d$ and hence $d \mid n \iff d \mid n+L$. The two sums therefore have identical summands. ∎

A Selberg-type sieve weight $n \mapsto \Sigma_{d\mid n,\ d\le D}\lambda_d$ is exactly of this form with $S = \{1,\dots,D\}$ and $L = \operatorname{lcm}(1,\dots,D)$. Hence **every bounded-support sieve weight is periodic**. Lean: `Periodic.divisor_sum_periodic`.

Lemma 5.2 (the twin indicator is not periodic)

PROOF

$\mathrm{twinInd}(3) = 1$ because 3 and 5 are both prime. For any $L \ge 1$, $3+3L = 3(1+L)$ is divisible by 3 and exceeds 3, hence composite, so $\mathrm{twinInd}(3+3L) = 0$. If $\mathrm{twinInd}$ had period L, iterating the period would give $\mathrm{twinInd}(3) = \mathrm{twinInd}(3+3L)$ — a contradiction. ∎

Lean: `NonPeriodic.twinInd_not_periodic`.

Theorem 5.3

$$\neg\ \exists\ (S \text{ finite})\ (L \ge 1)\ (\forall\ d\in S,\ d\mid L)\ (g : \mathbb{N} \to \mathbb{R}),\ \ \forall\ n,\ \ \mathrm{twinInd}\ n = \Sigma_{d\in S,\ d\mid n} g(d) .$$

PROOF

The right-hand side is L-periodic by Lemma 5.1; the left-hand side is not, by Lemma 5.2. ∎

Lean: `NonPeriodic.twinInd_not_divisor_sum`, on top of `Periodic` and `DivisorSum` (the latter also proves the multiplicative obstruction: a divisor sum $\zeta * g$ with multiplicative g is multiplicative, and the twin indicator is not).

> **Scope correction.** The synthesis states this as "complete for the entire Selberg class" (twice). That is stronger than what is proved. The theorem excludes *bounded-support, hence periodic, divisor sums* — i.e. classical Selberg sieve weights of finite level. The Selberg class is a class of Dirichlet series with an Euler product and a functional equation; nothing here rules out weights built from automorphic forms, and the sources say so elsewhere ("the missing object is a non-periodic weight"). The correct statement, and the one this harvest

reports, is: **the twin indicator is not a bounded-support divisor sum, so no classical finite-level Selberg sieve can produce it.** The quantitative strengthening — that the two-point count is off by a factor 2 — is numerical only.

FINDING 6 The negative map: what the programme ruled out

PROVED REFUTED A large part of the value here is subtractive. Each item below closes a door, and closes it with a computation or an argument rather than an intuition.

Results that constrain where the difficulty can live.

RESULT	CONTENT	EVIDENCE
The low-degree miracle stops at $d = 5$	Exact polynomial formulas hold for the generic twin count at $d = 3,4$ and fail at $d = 5$: no polynomial of any degree fits, by the $q^5/25$ degree bound plus interpolation error.	Errors at $q = 29,37,43$: $5.1\cdot10^4$, $5.5\cdot10^5$, $1.8\cdot10^6$
Value-set collapse for $d \geq 5$	The cap is a $d = 4$ phenomenon. For $d = 8,9$ at $q = 5$ the null polynomial $x^{d-q}(x^q-x)$ is extremal with $\lvert Im\rvert = 1$, and the maximum exceeds $(q-1)/2$.	Explicit extremal families, $q = 5,\ldots,9$
The j-line has density $1/4$	For $a_2 \neq 0$ the count depends only on $j = a_1^2/a_2^3$ and satisfies $I(j) = (q-1)/4 + O(1)$, fluctuation at most 5 — far below the Weil range $2\sqrt{q}$. Only the degenerate class x^4+b reaches density $1/2$: the cap is a cusp phenomenon.	$q = 13,29,37,41$
Refuted: the elliptic-trace probe	The j-line fluctuation was conjectured to be $-a_j/2$; it is not. Correlation $+0.30, +0.25, -0.10, -0.18$; identity holds at $0/13, 0/29, 1/37, 0/41$ shifts.	Round 30, direct test
Refuted: the class-number probe	The extremal count is not governed by an imaginary quadratic class number.	Round 22
Refuted: m as one trace	m cannot be a single Frobenius trace: at $q = 41$, $m = 13 > 2\sqrt{41}$.	Round 86

Item 1 is the most consequential. It says the exact-formula regime is genuinely low-degree: whatever structure produces $A = (3q^2+2q+3)/16$ at $d = 4$ does not survive one more degree. Any programme that hoped to climb d to a general theorem has to explain this wall.

FINDING 7 What was *not* achieved

For symmetry, the negative ledger. Stated plainly, so the map is not mistaken for a solution:

- **No proof of the twin prime conjecture**, over $\mathbb{Q}$ or over $\mathbb{F}_q[x]$. The function-field count is *cited* (Pollack; Bary-Soroker–Stix), not reproved; the fixed-q, $n \to \infty$ asymptotic remains conjectural in the literature the programme searched.
- **No crossing of the parity barrier.** Target C is an impossibility theorem: it says a class of methods fails. It does not construct the non-periodic weight that would succeed, and over $\mathbb{Q}$ that construction is the open problem.
- **No formalisation of the cap's generic case.** Ritt/Guralnick–Saxl, Poincaré reducibility and the S_4 decomposition are not in Lean. The certified layer covers the algebraic links and the finite base ($q \leq 23$), not the analytic input.
- **No proof of the last identity.** Five primes is not a theorem. The mechanism (twisted Parseval) is Probe 47, explicitly speculative.
- **No sharp quantitative parity constant.** The "factor 2" separation in Target C is numerical.
- **No structural explanation for $m = (a^2+1)/2$.** The formula is verified and identified with $J(\chi_4, \chi_4)$, but the derivation from the Gauss-sum expansion is not carried out.

3 The ledger

Every substantive claim in the programme, at its true status. This table is the answer to "what has been achieved", in one screen.

Status ledger. “Lean” = machine-checked; “Verified” = exact finite computation; “n primes” = number of primes at which the claim was checked.

CLAIM	STATUS	EVIDENCE / SOURCE
Target A — the d=4 cap	PROVED modulo 3 theorems	see F2
Translation $a_3 \mapsto 0$	LEAN	`Shift`
`R = (−Im g) ∪ Q`	LEAN	`Factorization`
Composition case `R = Q`	LEAN	`Composition`
x^4+b size	LEAN	`NegSquare`
$a_2 \neq 0$ size $q-\lfloor (q+2)/4 \rfloor$	LEAN	`QSize` , `QOrbit`
Generic bound $\lvert \mathrm{Im} \rvert \geq 5q/8 - \frac{1}{2}\sqrt{q} - O(1)$	CONDITIONAL	Ritt + Poincaré + Weil + S_4
Finite base `q < 49`	LEAN + CHECKED	`SmallCap` , Python
Target B — closed forms		
$T(a^d h) = T(h)$; constancy on power classes	PROVED	scaling, F3(a)
Amplitudes are a d-point DFT	PROVED	F3(b)
`T(d;q,h) = T(d;q,−h)`	LEAN	`TwinEven`
$q \mid T(d;q,h)$ for $d < q$	PROVED + LEAN	`FreeOrbit` ; sharp at d=q=5
$B(a_2)$ characterization and sizes	PROVED	F3(e)
$A = (3q^2+2q+3)/16$	PROVED	F3(f), 7 primes
$m = (a^2+1)/2$	28 PRIMES	F3(g)
$E(h)$ closed form, parity law	7 PRIMES	F3(h)
$\Sigma_{a_2} \lvert H \rvert^2$ formula	7 PRIMES	F4
Last identity (twisted Parseval)	80 PRIMES, EXACT / OPEN	F4, 019
Target C — parity		
Bounded-support divisor sums are periodic	PROVED + LEAN	`Periodic`

CLAIM	STATUS	EVIDENCE / SOURCE
Twin indicator is not periodic	PROVED + LEAN	`NonPeriodic`
Hence not a bounded-support divisor sum	PROVED + LEAN	`NonPeriodic`, `DivisorSum`
Separation theorem (zeta cannot see twins)	PROVED + LEAN	`Separation`, witness over $\mathbb{F}_7$
Quantitative factor 2	NUMERICAL	Target C, open
“Complete for the entire Selberg class”	OVERSTATED	see §7

4 Speculative probes

The programme keeps a live register of speculative mechanisms, each recorded with its evidence, the reason it is speculative, and a falsification test. They are called **probes** here — instruments sent ahead of the proof, expected to come back with either confirmation or failure — and the register is reproduced in full, because it is one of the more valuable parts of the record: a list of what the programme believed, what happened to those beliefs, and what would kill each one.

Probe register. Numbers are the sources' own.

#	PROBE	OUTCOME
3	The parity barrier *is* the periodicity gap; the missing object is a non-periodic weight (automorphic, $GL(2)\times GL(2)$-type).	LIVE The gap is proved; the identification with the barrier is an analogy.
17	Factor-set "compensation": at $d=4$ the quadratic condition Q compensates the value set; at $d \geq 5$ the factor sets are too small.	LIVE Consistent with equality cases; no proof of the mechanism.
23	The mod-8 congruence on $2B_4/q$ is the 2-adic weight of the orbit space (a Heisenberg-type extension).	LIVE Not identified.
28	m is a sum of finitely many Frobenius traces attached to the family $x^4+a_2x^2+b$.	LIVE Consistent with $m > 2\sqrt{q}$; number of terms unknown.
31	The Gauss-sum route to m: expand $\hat{B}(c)$, get $\zeta^{ca_2^2/4}$, hence a Gauss sum and the $\sqrt{q}$ term.	LIVE Now largely superseded by the $m=(a^2+1)/2$ identification.
32	$m = (a_C-3)/2 + c(q)$, a Fermat-quartic surface correction.	SUPERSEDED The partial match ($q=17,41,89$) was replaced by the exact $J(\chi_4,\chi_4)$ formula.
36	Proof of m by Gauss-sum expansion plus the classical $J(\chi_4,\chi_4)=\pm(a+bi)$.	LIVE The natural route; not carried out.
40	Kloosterman/Salié evaluation of $\Sigma_{a_2}\chi_2(a_2^2-4v)\chi_2(a_2^2-4w)$ after completing the square.	LIVE Concrete route to the last identity.
45	$K(A,c)$ is a conic point count; $\Sigma\lvert H\rvert^2$ follows from the conic/Weil bound.	LIVE Reformulation exact; evaluation missing.
46	$\Sigma_A\lvert K\rvert^2 = q^2-2q$ is Parseval; the twisted sum is χ_2-twisted Parseval giving J^2.	LIVE This is Probe 47's parent; the untwisted half is clean.
47	The last identity is twisted Parseval; then $m=(a^2+1)/2$ and the c_4 congruence are unconditional.	LIVE Closest to a proof strategy; still a probe.

Two entries in this register were tested and *died* — the elliptic-trace reading of the j-line (Probe of round 29, refuted in round 30) and the class-number reading (round 22). A register that records its own failures is more useful than one that does not; that is the main reason to keep the probes in a harvest rather than discarding them.

5 Significance, in plain terms

5.1 The one-sentence version

The programme did not get closer to proving the twin prime conjecture; it got *precise about why the obvious tools cannot*, and it built a small, fully checkable universe in which the failure can be measured.

5.2 The layman's analogy

Imagine primes as a row of lamps, and the twin prime conjecture as the claim that two adjacent lamps are lit infinitely often. The classical tool — the zeta function — is a machine that reads the row one lamp at a time and reports a total. It is very good at totals. But "two adjacent lamps are both lit" is a statement about a *pair*, and a machine that only totals single lamps cannot see it. Not because it is a weak machine, but because the information was never in its input.

That is what Finding 1 proves, and it is the reason Selberg's parity barrier is not a technical nuisance. The programme then did three useful things with the insight:

- **It built a miniature world where the question can be settled.** In $\mathbb{F}_q[x]$ the pair question has an answer, and the answer does not come from the zeta — it comes from a different mechanism (two-dimensional equidistribution). So the miniature world demonstrates both halves: the zeta is blind, and the blindness is not fatal *there* because another tool exists. Over the integers, that other tool does not exist. That is the whole story of the barrier in one picture.
- **It found a finite statement that is genuinely hard in the same way.** The $d=4$ cap says: for a quartic, the set of constants making it reducible is never more than half. This is a coin-flip-like bound, it is true, and its proof needs the same "compensation" that the parity barrier denies over $\mathbb{Q}$ — the quadratic-factor condition quietly filling the gap left by the value set. One can therefore *see* the mechanism that is missing upstairs, in a place where it is present and provable.
- **It drew the boundary of what its own method can reach.** Findings 5 and 6 say, with proof, that a whole family of approaches fails, and identify the one object that would have to exist for a success: a non-periodic weight with arithmetic content.

5.3 Why a negative result is the achievement

In this field a proof that a method *cannot* work is often worth more than another partial result, because it redirects effort. The programme's honest contribution is a map of a wall with a measured shape: which statistics each tool sees, which degree the exact-formula regime dies at, which finite-field inequality encodes the obstruction, and which single identity remains. A researcher who wants to attack the twin prime problem can use that map to avoid the dead ends without re-deriving them.

There is also an epistemic achievement, which is easy to miss: the programme kept a status register that distinguishes proved from certified from verified from guessed, and it recorded its own refutations. Most exploratory mathematics of this kind does not survive that discipline. Here the discipline is the product.

5.4 What would actually move the needle

1. **Prove the last identity.** It is one line, it now holds exactly at 80 primes in its corrected $\chi_2(2c)$ form, and it is the only thing between the closed forms and an unconditional c_4 chain. This is the highest-value, lowest-risk target in the programme — and 019 has already fixed the statement to aim at.
2. **Formalise Ritt + Poincaré + the S_4 decomposition.** This converts the cap from "proved modulo three standard theorems" to certified. It is engineering, not insight, and the rest of the certified layer shows it is feasible.
3. **Explain the d=5 wall.** Finding 6 says the exact-formula regime stops; a proof that it must stop would be a genuine structural theorem about twin counts over function fields.
4. **Sharpen the quantitative parity constant.** Turning the numerical factor 2 into a theorem would make Target C quantitative rather than merely qualitative.

6 Application to other problems

The results are specialised, but three of them transfer. The transfer is at the level of *technique and statement*, not of immediate corollaries; that is the honest scope.

6.1 Value sets and reducible specialisations of polynomials

The $d=4$ cap is a theorem about how a polynomial's value set interacts with its quadratic-factor condition. Statements of that shape — "at most/most half of the constants b make $f+b$ reducible" — are the finite-field engine behind counting reducible specialisations and, dually, counting irreducible values. The same circle of ideas feeds:

- **Function-field Bateman–Horn / Schinzel:** counting tuples of polynomials that are simultaneously irreducible is exactly the $T(d;q,h)$ problem, and any improvement in the exact formulas at fixed degree is a sharpening of the finite-field statement.
- **Value-set bounds:** the quantity $\max_f|\mathrm{Im}\ f|$ for degree-d polynomials is studied for its own sake (and interacts with permutation polynomials and with the "few-valued" polynomials used in cryptography). The programme's collapse results at $d \geq 5$ are data about exactly that quantity.
- **Coding and sequence design:** bounds on value sets over prime fields are used when a polynomial is required to have a large or small image; a sharp $(q+1)/2$ threshold with explicit equality families is

directly quotable.

6.2 Character sums, Jacobi sums, and the twisted-Parseval template

The remaining identity is an instance of a general and very common pattern: an untwisted second moment is clean (orthogonality/Parseval), and the twist by a quadratic character is the whole arithmetic content, evaluating to a Jacobi sum. The same template appears whenever one computes second moments of exponential sums with a character weight — in the analytic theory of L-functions, in the spectral statistics of Kloosterman/Salié sums, and in explicit computations of Frobenius traces on curves with extra automorphisms (the quartic/Fermat family here). A proof of the identity would be a worked example of the template with all constants pinned; the identification `m = (|Re J(χ₄,χ₄)|²+1)/2` is already a reusable identity.

6.3 Formalisation of negative results

The parity theorem and the separation theorem are rare specimens: **obstruction theorems that are machine-checked**. The Lean development supplies

- `FreeOrbit.card_dvd_card_of_free_action` — orbit–stabiliser plus class formula, a general divisibility engine;
- `Separation.no_zeta_observable` — the "a statistic is not a function of another statistic" pattern, with an explicit witness;
- `Periodic.divisor_sum_periodic` and `NonPeriodic.twinInd_not_divisor_sum` — the periodicity-versus-arithmetic obstruction, reusable for any indicator with a comparable rigidity;
- `SmallCap.cap_*` — `decide` theorems over `Finset.image`, a template for discharging finite ranges of a number-theoretic bound inside a proof assistant.

This is the most portable part of the programme: the statements are about arbitrary finite sets, groups and characters, so they can be instantiated far from twin primes.

6.4 What does *not* transfer

It should be said plainly: none of this yields a new bound toward the twin prime conjecture over $\mathbb{Z}$, and the function-field results do not imply anything over $\mathbb{Q}$ — the programme's own separation theorem explains why (there is no Frobenius to supply the pair equidistribution). Anyone citing this work for a twin-prime advance would be misreading it.

7 Caveats and corrections

A harvest that repeats its sources' overstatements is not a harvest. The following are places where the sources' language runs ahead of their evidence; each is corrected in this document.

Corrections applied in this harvest.

SOURCE CLAIM	PROBLEM	THIS DOCUMENT
"Target C: complete for the entire Selberg class."	What is proved is exclusion of bounded-support (periodic) divisor sums — the classical finite-level Selberg sieve class. Automorphic weights are not excluded; the sources say so elsewhere.	"Not a bounded-support divisor sum; no finite-level Selberg sieve can produce it." (§2 F5)
"The last identity … Then $m=(a^2+1)/2$ and the c_4 congruence is unconditional."	The antecedent is a probe, not a theorem; the consequent is conditional on it.	Reported as would-be-unconditional; status Open. (§2 F4)
"Proved modulo Ritt + Poincaré + the S_4 decomposition."	A precise conditional statement, but the synthesis's own remaining-work list includes formalising exactly those three items — and the programme's certified layer does not contain them.	Filed as *conditional*, flagged in §2 F2.
"All three are sharp: the computations of rounds 1–106 show no further structural simplification."	An empirical claim about the absence of structure. It is evidence about a search, not a theorem.	Reported as a programme judgement, not a result. (§5)
Document header: "Round 50", in a file named round106, citing rounds 62–66 and 1–106.	The synthesis is assembled from earlier ones and its round numbering is inconsistent. It is the third "final synthesis" (after rounds 79 and 88), with essentially the same residual gap.	Cited as the round-106 state; the churn is noted. (§10)
"$q \mid T(d;q,h)$", with the action described as $t\cdot(f,g) = (f+t,\ g+t)$.	That operation does not preserve irreducibility. The preserved operation is $f(X) \mapsto f(X+a)$, with freeness requiring $d < q$.	Correct action stated, with the $d=q=5$ counterexample. (§2 F3(c))
m and $E(h)$ formulas stated without hypotheses.	They presuppose $q \equiv 1 \pmod 8$ (odd a, $b \equiv 0 \bmod 4$) for the canonical two-square representation, and all the data are at such primes.	Hypothesis stated explicitly. (§2 F3(g),(h))
"Verified at 5 primes" presented adjacent to proved items.	A finite verification is not a proof and should not share a status column with one.	Separate status tags throughout; ledger in §3.

> **A note on the name change.** The sources' own word for the speculative register is replaced by **probe** throughout, and the substitution is not cosmetic: a probe is an instrument you send ahead of the proof to find out what is there, and it is expected to come back with either confirmation or failure. Two of the programme's probes did fail, and the register records it. That is the intended use.

8 Glossary

Qrime	A monic irreducible polynomial over $\mathbb{F}_q$ — the polynomial-model analogue of a prime. “Qrime” is the programme's coinage.
Twin qrime	A pair `(f, f+2)` of qrimes of the same degree. The count of such `f` is `T(d;q,h)` with `h = 2`.
World	A set `S` of qrimes with a degree function. Quantifying over worlds (rather than only over the true one) is what makes the separation theorem a statement about information.
Zeta datum	The degree-count sequence `Z(S)(n) = #{f ∈ S : deg f = n}`. By the Euler product and Möbius inversion this is exactly what ζ_S sees.
Linear / bilinear statistic	Linear: a sum of one-point indicators (the qrime count, zeta coefficients, Frobenius traces). Bilinear: a sum of products of two indicators at a related pair (the twin count). The zeta sees the first, not the second.
Value set `Im(g)`	The image of `g : 𝔽q → 𝔽q`. Its size controls the linear-factor condition.
Quadratic-factor set `Q`	The set of constants `b` for which `g+b` has a quadratic factor; explicitly `Q = { ((a₂+p²)² − (a₁/p)²)/4 : p ≠ 0 }` plus a degenerate term.
The cap	The inequality `|(−Im g) ∪ Q| ≥ (q+1)/2` for quartics, equivalently `I(a) ≤ (q−1)/2`. Target A.
Monodromy	The action of the Galois group of the splitting field of `g(x) − t` on the fibre. Primitivity of this action is what Ritt's theorem gives for a non-composition, and it forces the value set to be large.
Ritt's theorem	Structure theorem for polynomial decompositions: a polynomial is a composition exactly when its monodromy is imprimitive (Guralnick–Saxl form). Cited, not

	formalised.
Poincaré reducibility	A surjection of abelian varieties with connected fibres splits up to isogeny; used here as $\mathrm{Jac}(C) \sim E \times E'$ for the genus-2 curve $C : s^2 = S(x)$.
Weil bound	$\lvert a_E \rvert \le 2\sqrt{q}$ for the Frobenius trace of an elliptic curve; the source of the $O(\sqrt{q})$ errors here.
Class function / S_4 decomposition	Expansion of a fibre indicator in the irreducible characters of S_4. The 2-dimensional irreducible appears as χ_2 here — not to be confused with the quadratic character of $\mathbb{F}_q$.
DFT (discrete Fourier transform)	Since T is constant on d-th power classes, it is a function on $\mathbb{Z}/d$, and its Fourier amplitudes $A_c = \Sigma_k T_k \zeta_d^{ck}$ are the useful coordinates.
Jacobi sum $J(\chi_4,\chi_4)$	The classical quartic Jacobi sum, evaluating to $\pm(a+bi)$ when $q = a^2+b^2$. It carries the constant m.
Selberg class / sieve weight	A Dirichlet series with Euler product and functional equation; a sieve weight is $n \mapsto \Sigma_{d\mid n, d\le D}\lambda_d$. The latter is periodic — the fact that powers the parity theorem.
Parity barrier	Selberg's obstruction: sieve-type (linear) methods cannot detect the parity of $\Omega(n)$, so they cannot force a two-point correlation. The programme's finite form: periodic weights cannot equal the twin indicator.
Probe	A speculative mechanism recorded with its evidence, its reason for being speculative, and a falsification test. See §4.
Lean / Mathlib	Lean 4 is a proof assistant; Mathlib is its mathematics library. “Lean-certified” means the statement is checked by the kernel, not that the informal reading has been verified.

9 Citations

Items 1–8 are the external theorems the programme relies on or quotes; 9–13 are the programme's own internal sources, given for traceability. The arXiv identifiers are the ones cited in the programme's notes.

1. J. F. Ritt, *Prime and composite polynomials*, Trans. Amer. Math. Soc. **23** (1922), 51–66. — the decomposition/primitivity theorem used in step (4) of the cap.

2. R. Guralnick and J. Saxl, *Monodromy groups of polynomials*, in *Galois Groups and Fundamental Groups*, MSRI Publ. **41** (2003), 125–150. — the form of Ritt used here: non-composition ⟹ primitive monodromy.

3. H. Poincaré, *Sur les propriétés arithmétiques des courbes algébriques*, J. Math. Pures Appl. **7** (1901), 161–233. — complete reducibility of abelian varieties, used via $\mathrm{Jac}(C) \sim E \times E'$.

4. A. Weil, *On some exponential sums*, Proc. Nat. Acad. Sci. USA **34** (1948), 204–207; and *Numbers of solutions of equations in finite fields*, Bull. Amer. Math. Soc. **55** (1949), 497–508. — the bound $|a_E| \le 2\sqrt{q}$ and the character-sum estimates.

5. A. Selberg, *On elementary methods in prime number theory and their limitations*, Den 11te skandinaviske matematikerkongress (1952), 13–22. — the parity phenomenon that Target C formalises in finite form.

6. P. Pollack, function-field Bateman–Horn: Conjecture 2 and Theorem A, as cited in notes/round2.md and notes/round3-formal-statement.md. — the asymptotic $T(n) \sim S(q) q^n / n^2$ is cited, not reproved; the large-q regime is Pollack's theorem, the fixed-q case is conjectural in the literature searched.

7. L. Bary-Soroker, *Irreducible values of polynomials*, Adv. Math. **229** (2012), 854–874; arXiv:1005.4528. — explicit statement that the $q^n \to \infty$ formula is conjectural.

8. L. Bary-Soroker and J. Stix, arXiv:1711.05564, Thm. 1.2 and 1.4. — the exact degree-3 twin-qrime formula, verified against the sieve in notes/round6-exact-formula.md.

9. T. Tao, *The logarithmically averaged Chowla and Sarnak conjectures are equivalent*, arXiv:1605.04628. — $\lambda = 1_{\text{squares}} * \mu$, hence one-point Chowla follows from $M(y) = o(y)$; used in the parent programme's corrections.

10. The Lean 4 proof assistant and Mathlib4, leanprover-community.github.io/mathlib4. — engine of the certified layer (45 modules in `018/lean`).

11. H. Iwaniec and E. Kowalski, *Analytic Number Theory*, AMS Colloq. Publ. **53** (2004). — background for Selberg sieve weights, Weil bounds and exponential-sum technology.

12. K. Ireland and M. Rosen, *A Classical Introduction to Modern Number Theory*, 2nd ed., GTM **84**, Springer (1990), ch. 8–9. — quadratic/quartic characters, Gauss and Jacobi sums, including $J(\chi_4, \chi_4) = \pm(a+bi)$.

13. Internal sources: 018/README.md, notes/round106-final-synthesis-v2.md, notes/cap-proof.md, notes/round3-formal-statement.md, notes/round58-parity-theorem.md, notes/round89-B-characterization.md, notes/round94-m-closed.md, notes/round95-jacobi.md, notes/round99-Hsum.md, notes/round105-Ksums.md. — the 106 notes and 111 check scripts in `018/notes` and `018/checks` are the primary record.

10 Reproduction and provenance

Every numerical claim in this harvest is attached to an executable script, and every algebraic claim to a Lean module or a note. The layout:

```
research/twin-prime-parity/
├── 018/                  the source programme (read-only here)
│   ├── notes/            106 round notes (the primary record)
│   ├── lean/              45 Lean modules + check.sh
│   ├── checks/           111 Python verification scripts
│   └── lib/               shared helpers (e.g. singular series)
└── 019/                  this document + the F4 falsification test
    ├── harvest.html      this document
    ├── README.md         the 019 addendum
    ├── lib/twisted_parseval.py       exact arithmetic in Z[zeta_q]
    ├── checks/falsification_test.py  80 primes, exact
    ├── checks/sympy_crosscheck.py    independent sympy/mpmath path
    ├── tests/test_twisted_parseval.py  14 unit tests
    ├── lean/Twist.lean   the certified splitting lemma
    └── results.json      the raw test rows
```

Reproducing the certified layer

```
cd research/twin-prime-parity/018/lean
bash check.sh *.lean          # requires $HOME/lean with Mathlib4
# -> [check] OK  for each module, exit 0
```

VERIFIED AT HARVEST TIME The sweep was actually run while preparing this document: **all 45 modules re-elaborate clean, exit 0**, and the checked statements report only the three standard axioms (`propext`, `Classical.choice`, `Quot.sound`) — no `sorryAx`, i.e. no unproved holes in the certified layer. The one diagnostic is a style warning in `Scale.lean` (`haveI` versus `have`), which does not affect any statement.

> **What that does and does not certify.** It certifies the *statements* in `018/lean`: the translation reduction, the union decomposition, the composition case, the orbit counts, the finite base, the periodicity pair, the separation witness, and the abstract free-orbit engine. It does *not* certify the generic case of the cap (§2 F2), because Ritt, Poincaré reducibility and the S_4 decomposition are not in the module set. The distinction between “the Lean layer is green” and “the cap is proved” is exactly the distinction this harvest is trying to preserve.

Reproducing the quantitative layer

```
python3 checks/round89_B_characterization.py  # B(a2) sizes
python3 checks/round94_m_closed.py            # m = (a^2+1)/2 at many primes
python3 checks/round95_jacobi.py              # J(chi4,chi4) values
python3 checks/round105_Ksums.py              # the two K-sums
python3 checks/round106_sw.py                 # the latest sweep
```

Script names as listed in `018/checks`; each round note carries its own reproduction line at the end. The certified layer is designed to re-elaborate from scratch against a local Mathlib toolchain; the module headers record which round produced each file, and `check.sh` is the same harness used by the parent programme's rounds.

Reproducing the 019 addendum

```
cd research/twin-prime-parity/019
python3 checks/falsification_test.py          # 80 primes, exact; writes results.json
python3 checks/sympy_crosscheck.py            # sympy / mpmath, 50 digits
python3 -m unittest discover -s tests -v      # 14 unit tests
cd lean && ./check.sh Twist.lean Verify.lean  # the certified splitting lemma
```

The exact test is a proof for each prime tested: a pass means the two sides are the same element of $\mathbb{Z}[\zeta_q]$, not that they agree numerically. The failure of the strict form at $q \equiv 5 \pmod 8$ is likewise exact — an integer vector inequality, not a rounding effect.

> **Provenance.** This document is a harvest — a re-reading and re-grading — of notes/round106-final-synthesis-v2.md and the notes it cites. It was written at round 106, after the aggressive goal, and it deliberately re-grades several of the sources' status claims (§7). Where this document and the sources disagree, the sources are the primary record and this document's status column is the judgement; the disagreements are itemised rather than smoothed over.

> **Moved and extended.** The document was first written in `018/`; it now lives in `019/` together with the completed falsification test of §2 F4. The round-106 sources are unchanged and are linked from here. The two substantive additions are the 75 new primes and the corrected $\chi_2(2c)$ form, and the Lean proof of the splitting step.

No twin prime conjecture was proved or refuted in the making of this document. Speculative items are labelled probes; verified items are labelled verified; the ledger in §3 is the compressed statement of what is actually known.